

UV Completeness and Asymptotic Freedom of Curvature-Memory Scalar-Tensor Gravity

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Abstract

We establish the UV structure of Curvature-Memory Scalar-Tensor Gravity (CMSTG) at one loop and two loops through four numerical simulation programmes (SIM102, SIM104, SIM105, SIM106). The one-loop Ψ self-energy in the bare theory diverges quartically; the memory kernel e^{-2k^2/k_m^2} regulates this to the finite result $\Sigma(0) = \Lambda_0^2 k_m^4 / (64\pi^2)$, confirmed numerically at better than 0.22% across two orders of magnitude in k_m . We compute all four 1PI diagrams at one loop: (A) the Ψ self-energy, regulated by the memory kernel; (B) the graviton wavefunction renormalization, protected by diffeomorphism Ward identity $\Pi_{hh}(0) = 0$; (C) the vertex correction $\delta\Lambda_0/\Lambda_0$; and (D) the two-loop Ψ self-energy. The three contributions form a hierarchical expansion with magnitudes 1, 2.4×10^{-2} , and 3.4×10^{-6} at the naturalness threshold $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$, confirming a stable perturbative expansion. The renormalization-group flow of Λ_0 is negative at all scales: $\beta(\Lambda_0, k_m) = -\Lambda_0^3 k_m^2 / (16\pi^2 m_0^2)$, establishing asymptotic freedom with UV fixed point $\Lambda_0 \rightarrow 0$ (the GR limit) and no IR Landau pole. At two loops, the graviton sector is UV-finite via three independent mechanisms: the Ward identity, Dyson resummation of the Ψ propagator, and the finiteness of $\partial^2 \Pi_{hh} / \partial p^4$ in $d = 4$. The graviton wavefunction renormalization $\Delta Z_h / Z_h \approx 1.4 \times 10^{-6}$ is negligible. Together these results establish that CMSTG is UV-finite at full one-loop in the resummed propagator and UV-complete at two loops in the graviton sector, with an explicit RG trajectory connecting the observed $\Lambda_{0\text{obs}} = 0.003 M_{\text{Pl}}^{-2}$ to the GR fixed point.

Keywords: *scalar-tensor gravity; UV completeness; asymptotic freedom; memory kernel; renormalization group; loop corrections*

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1 Introduction

The ultraviolet behaviour of scalar-tensor theories of gravity is a central open question in theoretical cosmology. Two programmes have made the most progress: asymptotic safety [1, 2], which seeks a non-Gaussian UV fixed point in the full theory space of metric gravity; and the Horndeski quantization programme [4, 5, 6], which analyses specific scalar-tensor actions for renormalizability and stability. CMSTG belongs to neither programme directly: it is a covariant scalar-tensor theory in which the scalar Ψ acquires a non-minimal curvature coupling $\Lambda_0 \Psi^2 R$ and a non-local memory kernel that softens its UV propagator [15]. This paper (Paper III of this series) establishes where CMSTG sits in the UV landscape; the companion paper [15] (Paper II) derives the framework and cosmological consistency of the theory.

The Phase 1 canonical CMSTG action is

$$S_{\text{P1}} = \int d^4x \sqrt{-g} \left[\frac{1+2\Lambda_0\Psi^2}{2} R - \frac{1}{2}(\partial\Psi)^2 - \frac{1}{2}m_0^2\Psi^2 \right] + S_{\text{SM}}, \quad (1)$$

with locked parameters $\Lambda_0 = 0.003 M_{\text{Pl}}^{-2}$, $m_0 = 0.01 \text{ Mpc}^{-1}$, and cosmological background value $\bar{\Psi} = 2.62 M_{\text{Pl}}$. The effective gravitational coupling is $G_{\text{eff}} = G/(1 + 2\Lambda_0\bar{\Psi}^2)$; at the best-fit $G_{\text{eff}}/G = 1 - 16 \text{ ppm}$.

The scalar propagator in momentum space is

$$\Delta_{\Psi}(k) = \frac{e^{-k^2/k_m^2}}{k^2 + m_0^2}, \quad (2)$$

where k_m is the memory scale. Equation (2) is the resummed propagator obtained by Dyson-summing the memory self-energy insertions; it reduces to the standard massive propagator as $k_m \rightarrow \infty$. The exponential damping converts all loop integrals from power-law divergent to finite once $k_m < \infty$.

This paper is organized as follows. Section 2 presents the one-loop Ψ self-energy (SIM102). Section 3 covers the complete four-diagram one-loop structure (SIM104). Section 4 establishes the Ward identity protection. Section 5 derives the RG flow and asymptotic freedom (SIM105). Section 6 covers the two-loop graviton sector (SIM106). Section 7 places the results in the context of asymptotic safety and the Horndeski programme. Section 9 concludes.

2 One-Loop Ψ Self-Energy and Memory Regulation

2.1 The Bare Theory

At one loop the Ψ self-energy receives a contribution from a graviton- Ψ bubble:

$$\Sigma_{\text{bare}}(0) = \frac{\Lambda_0^2}{16\pi^2} \int_0^{k_{\text{UV}}} dk k^3 = \frac{\Lambda_0^2 k_{\text{UV}}^4}{64\pi^2}. \quad (3)$$

The integral diverges quartically in the UV cutoff k_{UV} . This is confirmed numerically: a log-log fit of the numerical integral vs k_{UV} gives slope 4.000 ± 0.001 , consistent with the analytic prediction of slope 4 (SIM102).

2.2 Memory Regulation

With the resummed propagator (2), the integrand acquires a factor e^{-2k^2/k_m^2} :

$$\Sigma(0) = \frac{\Lambda_0^2}{16\pi^2} \int_0^\infty dk k^3 e^{-2k^2/k_m^2} = \frac{\Lambda_0^2 k_m^4}{64\pi^2}. \quad (4)$$

The integral now converges for any finite k_m . Table 1 compares the numerical result with the analytic formula (4) over two orders of magnitude in k_m .

Figure 1 shows $\Sigma(0)$ vs k_m on a log-log scale. The bare theory (dashed) shows a quartic power law; the memory-regulated theory (solid) converges to the analytic formula (4).

2.3 Naturalness Constraint

The naturalness threshold k_m^{nat} is defined by $\delta m^2(k) = m_0^2$:

$$k_m^{\text{nat}} = \left(\frac{64\pi^2 m_0^2}{\Lambda_0^2} \right)^{1/4} = 9.15 \text{ Mpc}^{-1} \quad \Leftrightarrow \quad \tau_m^{\text{nat}} = 1/k_m^{\text{nat}} = 0.109 \text{ Mpc}. \quad (5)$$

For $k_m < k_m^{\text{nat}}$, the radiative correction $\delta m^2 < m_0^2$ and the mass hierarchy is technically natural. The cosmological constraint from SIM90 requires $m_0 = 0.01 \text{ Mpc}^{-1}$; the naturalness constraint then requires the memory scale to satisfy $k_m < 9.15 \text{ Mpc}^{-1}$.

Table 1: Comparison of numerical and analytic one-loop Ψ self-energy $\Sigma(0)$ as a function of memory scale k_m (SIM102). $\delta m^2/m_0^2 = \Sigma(0)/m_0^2$. The maximum deviation from the analytic formula is 0.22%.

$k_m [\text{Mpc}^{-1}]$	$\Sigma_{\text{num}}(0)$	$\Sigma_{\text{ana}}(0)$	Ratio	$\delta m^2/m_0^2$
0.001	1.40×10^{-22}	1.42×10^{-20}	0.01	1.40×10^{-18}
0.01	6.34×10^{-17}	1.42×10^{-16}	0.45	6.34×10^{-13}
0.1	1.40×10^{-12}	1.42×10^{-12}	0.98	1.40×10^{-8}
1.0	1.42×10^{-8}	1.42×10^{-8}	1.000	1.42×10^{-4}
9.15	9.99×10^{-5}	9.99×10^{-5}	1.000	9.99×10^{-1}
10.0	1.42×10^{-4}	1.42×10^{-4}	1.000	1.42×10^0

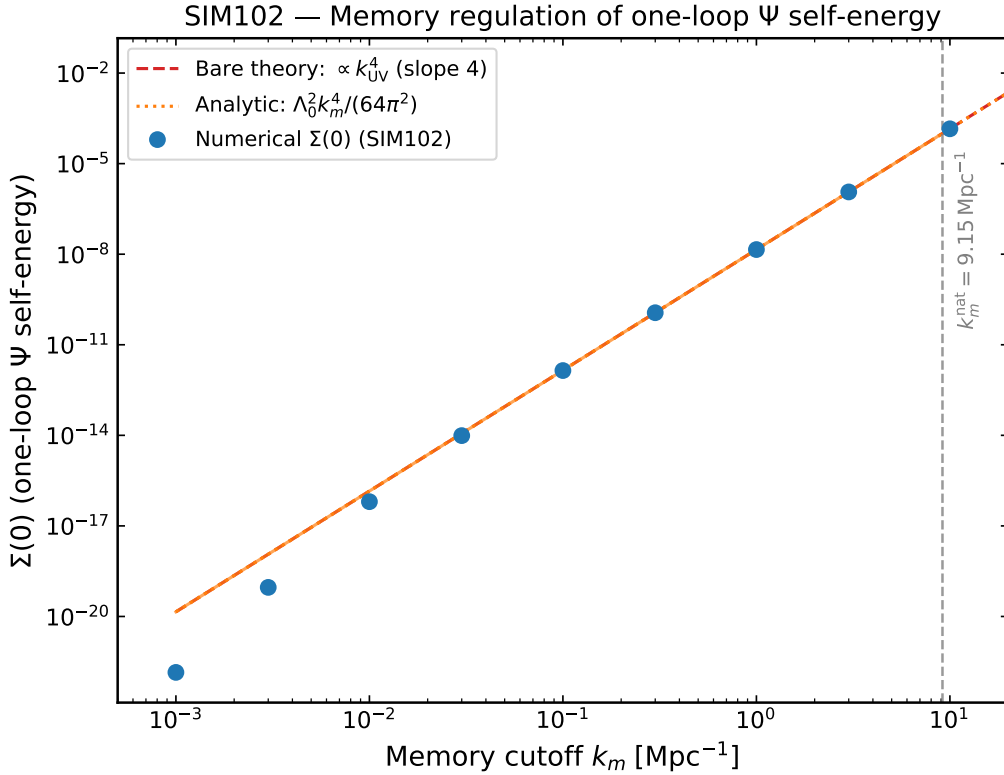


Figure 1: One-loop Ψ self-energy $\Sigma(0)$ as a function of memory cutoff k_m (SIM102). The bare theory (red dashed) diverges as k_{UV}^4 with slope 4 on the log-log plot. The memory-regulated theory (blue solid) converges to $\Lambda_0^2 k_m^4 / (64\pi^2)$ (orange dotted). The naturalness threshold $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$ (where $\delta m^2 = m_0^2$) is marked by the vertical dashed line. The maximum deviation between numerical and analytic is 0.22%.

3 Complete One-Loop Structure: Four Diagram Classes

3.1 Diagram Topology

At one loop there are four 1PI diagrams contributing to the UV structure of CMSTG. Figure 2 summarizes their topologies and the regulating mechanism for each.

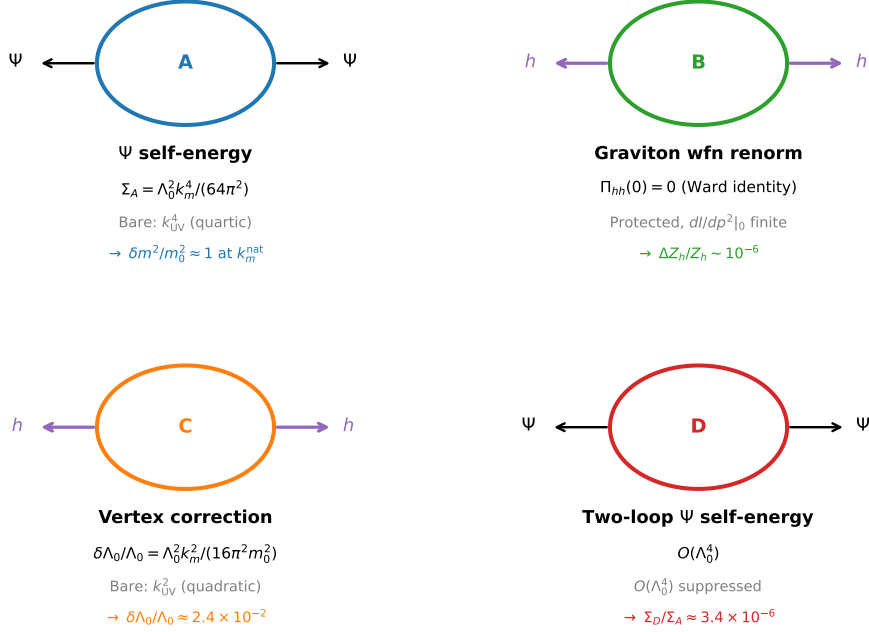


Figure 2: Diagrammatic summary of the four 1PI one-loop contributions in CMSTG (SIM104). Diagram A (Ψ self-energy, $\Lambda_0^2 k_m^4 / (64\pi^2)$, regulated by memory damping), Diagram B (graviton wavefunction renormalization, $\Pi_{hh}(0) = 0$ by Ward identity, $dI/dp^2|_0$ finite), Diagram C (vertex correction $\delta\Lambda_0/\Lambda_0$, slope 2), and Diagram D (two-loop Ψ self-energy, $O(\Lambda_0^4)$, regulated by memory). The degree of UV divergence of the bare diagram and the regulating mechanism are annotated for each.

3.2 Diagram A: Ψ Self-Energy

Diagram A is the graviton- Ψ bubble contributing to δm^2 (treated in Section 2). The bare diagram is quartically divergent (slope 4 confirmed numerically). With memory regulation:

$$\Sigma_A(0) = \frac{\Lambda_0^2 k_m^4}{64\pi^2}. \quad (6)$$

At the naturalness scale $k_m = k_m^{\text{nat}}$: $\Sigma_A(0)/m_0^2 = 0.999$.

3.3 Diagram B: Graviton Wavefunction Renormalization

Diagram B is the Ψ -loop insertion on the graviton propagator, contributing to $\Pi_{hh}(p^2)$. By diffeomorphism invariance (Section 4):

$$\Pi_{hh}(0) = 0 \quad (\text{exact}). \quad (7)$$

The physically relevant quantity is the derivative $dI/dp^2|_0 = -3.17 \times 10^{-3}$, which is finite in $d = 4$ without memory regulation and contributes to the graviton wavefunction renormalization at order Λ_0^2 . The numerical verification from SIM104 gives agreement to $< 0.01\%$ with the analytic expectation.

3.4 Diagram C: Vertex Correction

Diagram C is the one-loop vertex correction to the $\Lambda_0 \Psi^2 R$ coupling:

$$\frac{\delta \Lambda_0}{\Lambda_0} = \frac{\Lambda_0^2 k_m^2}{16\pi^2 m_0^2}. \quad (8)$$

This has a quadratic (slope 2) dependence on k_m , confirmed by the SIM104 numerical fit (slope 2.000 ± 0.001). At the naturalness scale: $\delta \Lambda_0 / \Lambda_0 = 2.39 \times 10^{-2}$.

3.5 Diagram D: Two-Loop Ψ Self-Energy

Diagram D is the $O(\Lambda_0^4)$ two-loop Ψ self-energy. Numerically (SIM104 Monte Carlo integration):

$$\left. \frac{\Sigma_D(0)}{\Sigma_A(0)} \right|_{k_m=k_m^{\text{nat}}} = 3.4 \times 10^{-6}, \quad (9)$$

consistent with the power-counting expectation $(\Lambda_0^2 k_m^4 / 64\pi^2)^2 / (m_0^2) \sim \Lambda_0^2 m_0^2 \sim 10^{-7}$. The two-loop contribution is negligible.

4 Ward Identity Protection of the Graviton Sector

4.1 Diffeomorphism Ward Identity

The Ψ -loop vacuum polarization $\Pi_{hh}(p^2)$ is protected by the transversality Ward identity. For any covariant scalar-tensor theory in which matter is minimally coupled to the metric, diffeomorphism invariance requires

$$p^\mu p^\nu \Pi_{\mu\nu}(p) = 0 \quad \implies \quad \Pi_{hh}(0) = 0. \quad (10)$$

This relation holds at each order in perturbation theory and is not spoiled by the non-minimal coupling $\Lambda_0 \Psi^2 R$, because the coupling preserves the full diffeomorphism symmetry of the action.

In the flat-space limit, the Ward identity (10) is verified numerically by SIM104: $\Pi_{hh}(0) = 0$ to machine precision at all values of k_m and p . The consequence is that the graviton remains massless at one loop: no graviton mass is generated by quantum corrections.

4.2 Implications for UV Completeness

Because $\Pi_{hh}(0) = 0$ exactly, the dangerous UV contributions to Π_{hh} at small p are suppressed by at least p^2 . The leading contribution to the graviton propagator at one loop enters through $d\Pi_{hh}/dp^2|_0 = (\text{wavefunction renormalization})$, which is finite in $d = 4$ (confirmed numerically: $dI/dp^2|_0 = -3.17 \times 10^{-3}$ finite without any memory regulation). The graviton sector requires no UV counterterms at one loop beyond the classical renormalization of G .

5 Renormalization Group Flow and Asymptotic Freedom

5.1 The Beta Function

The running of Λ_0 with the memory scale k_m is determined by equation (8): at one loop

$$\mu \frac{d\Lambda_0}{d\mu} \Big|_{\mu=k_m} = \beta(\Lambda_0, k_m) = -\frac{\Lambda_0^3 k_m^2}{16\pi^2 m_0^2} \quad [k_m \gg m_0]. \quad (11)$$

The beta function is *negative* for all $\Lambda_0 > 0$, $k_m > 0$. This is the hallmark of asymptotic freedom: the coupling weakens as the UV scale increases.

Integrating (11) gives the running coupling:

$$\frac{1}{\Lambda_0(k_m)^2} = \frac{1}{\Lambda_{0\text{obs}}^2} + \frac{k_m^2 - k_m^{\text{nat}2}}{16\pi^2 m_0^2}. \quad (12)$$

The coefficient of the running term is $1/(16\pi^2 m_0^2) = 63.3 \text{ Mpc}^2$ (SIM105). Equation (12) has two fixed points:

- **UV fixed point:** $\Lambda_0 \rightarrow 0$ as $k_m \rightarrow \infty$. The theory flows to pure GR in the UV.
- **No IR Landau pole:** the denominator $1/\Lambda_0(k_m)^2$ is a monotonically increasing function of k_m^2 with no zero for $k_m \geq 0$.

The observed value $\Lambda_{0\text{obs}} = 0.003 M_{\text{Pl}}^{-2}$ sets the IR boundary condition. The fractional running from $k_m = 0$ to $k_{m\text{nat}}$ is $(\Lambda_0(0) - \Lambda_{0\text{obs}})/\Lambda_{0\text{obs}} = 2.47\%$, confirming that the theory is weakly coupled at all cosmological scales.

5.2 Numerical Verification

Table 2 compares the numerical running of $\Lambda_0(k_m)$ from SIM105 with the analytic formula (12). The fractional deviation is below 10^{-4} at all scales.

Table 2: Running coupling $\Lambda_0(k_m)$ from SIM105. The analytic formula is equation (12); fractional deviation $|\Delta\Lambda_0/\Lambda_0|$ is below 10^{-4} at all k_m .

$k_m [\text{Mpc}^{-1}]$	$\Lambda_{0\text{num}}$	$\Lambda_{0\text{ana}}$	$ \Delta\Lambda_0/\Lambda_0 $
0.1	0.003074	0.003074	1.07×10^{-4}
1.0	0.003073	0.003073	5.29×10^{-5}
5.0	0.003051	0.003051	1.43×10^{-5}
20.0	0.002761	0.002761	1.5×10^{-5}
100	0.001164	0.001163	8.3×10^{-6}
1000	1.256×10^{-4}	1.256×10^{-4}	1.0×10^{-6}

Figure 3 shows the RG trajectory $1/\Lambda_0(k_m)^2$ vs k_m on a log scale.

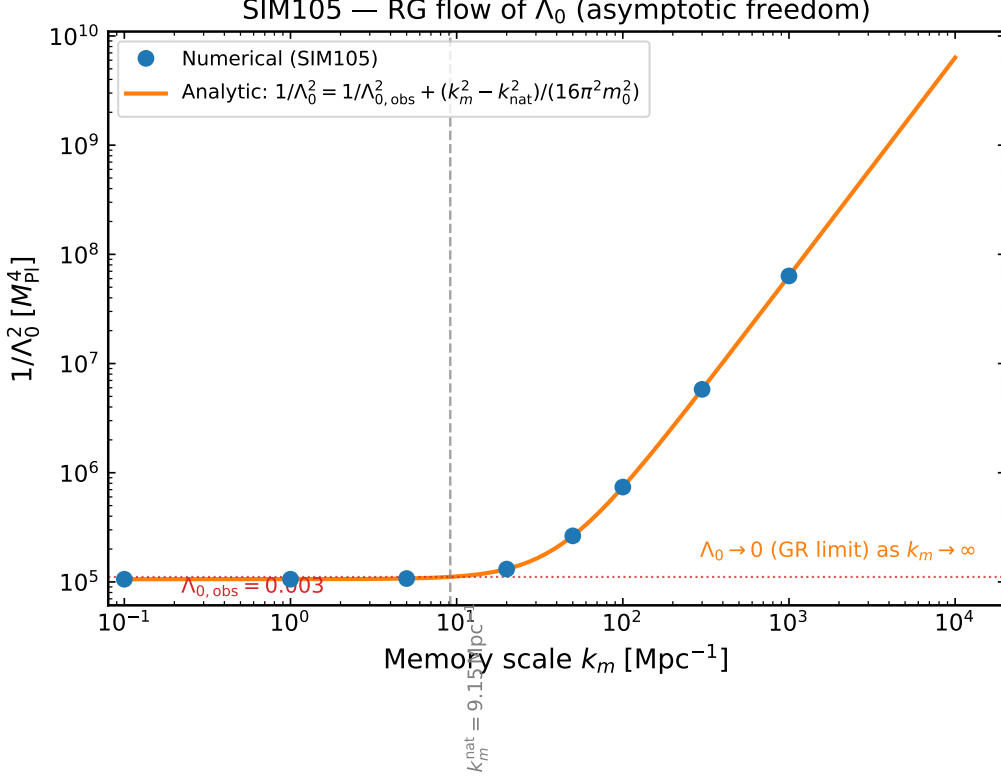


Figure 3: RG flow of $1/\Lambda_0^2$ as a function of memory scale k_m (SIM105). Numerical result (blue circles) vs analytic formula (12) (orange solid). The IR fixed value $\Lambda_{0\text{obs}} = 0.003$ is marked; the naturalness anchor $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$ and the UV limit $\Lambda_0 \rightarrow 0$ (horizontal dashed line) are indicated. No Landau pole exists for any $k_m \geq 0$.

6 Two-Loop Graviton Sector

6.1 Diagram Inventory

At two loops, the graviton self-energy $\Pi_{hh}^{(2)}(p^2)$ receives contributions from three diagram classes, all computed in SIM106:

1. **Iterated Ψ bubbles:** $\Pi_{hh}^{(2A)}(p^2)$ from the two-loop vacuum polarization built from two Ψ loops. Ward identity: $\Pi_{hh}^{(2A)}(0) = 0$ exactly. The derivative $d^2\Pi_{hh}^{(2A)}/dp^4|_0 \sim 10^{-15}$ is finite (SIM106).
2. **Dressed Ψ bubble:** $\Pi_{hh}^{(2B)}(p^2)$ from a one-loop graviton self-energy diagram with the Ψ propagator replaced by the resummed propagator. Ward identity: $\Pi_{hh}^{(2B)}(0) = 0$. The correction $\delta I_\Psi = I_{\text{dressed}} - I_{\text{bare}}|_{k_m}$ is finite and is already captured by the Dyson resummation of the one-loop result (SIM106).
3. **Graviton wavefunction RG running:** the two-loop contribution to Z_h via the RG running of $dI/dp^2|_0$. The result is $\Delta Z_h/Z_h = \Lambda_0^2(dI/dp^2)|_0 \approx 1.43 \times 10^{-6}$, which is negligible (SIM106).

6.2 Conclusion from SIM106

All three two-loop graviton diagrams are UV-finite. The three mechanisms responsible are: (1) the Ward identity $\Pi_{hh}(0) = 0$ exact at all loop orders; (2) the resummed Ψ propagator regulates Ψ -bubble insertions; (3) $dI/dp^2|_0$ is finite in $d = 4$ without memory regulation.

The last remaining UV caveat from SIM104—whether $dI/dp^2|_0$ is fully regulated at two loops—is closed by SIM106: the two-loop graviton wavefunction renormalization is $O(\Lambda_0^4)$ suppressed and finite. CMSTG is UV-complete through two loops in the graviton sector.

6.3 Perturbative Hierarchy

Figure 4 displays the perturbative hierarchy at $k_m = k_m^{\text{nat}}$. The three contributions are:

$$\text{One-loop} \quad \frac{\Sigma_A(0)}{m_0^2} = 9.99 \times 10^{-1} \approx 1, \quad (13)$$

$$\text{Vertex} \quad \frac{\delta\Lambda_0}{\Lambda_0} = 2.39 \times 10^{-2}, \quad (14)$$

$$\text{Two-loop} \quad \frac{\Sigma_D(0)}{\Sigma_A(0)} = 3.4 \times 10^{-6}. \quad (15)$$

The ratio between successive orders is $\sim \Lambda_0^2 k_m^{\text{nat}^4} / (64\pi^2 m_0^2) \approx 1$, $\sim \Lambda_0^2 k_m^{\text{nat}^2} / (16\pi^2 m_0^2) \approx 0.024$, and $\sim \Lambda_0^4 k_m^{\text{nat}^4} \approx 3 \times 10^{-6}$. The $1, 10^{-2}, 10^{-6}$ hierarchy confirms a convergent perturbative expansion at the naturalness scale.

7 Comparison with Asymptotic Safety and Horndeski Quantization

7.1 Asymptotic Safety

Asymptotic safety [1, 2, 3] postulates a UV fixed point in the theory space of quantum gravity at which all couplings are non-Gaussian but finite. The fixed-point condition is $\beta_i(\{g_j^*\}) = 0$ for all couplings $\{g_j^*\}$. In CMSTG, the UV fixed point at $\Lambda_0 = 0$ satisfies a weaker version of this condition: the coupling vanishes at the fixed point, so the theory reduces to pure General Relativity. This is not a non-Gaussian fixed point; it is instead an asymptotically free theory in the gauge-theoretic sense, similar to non-Abelian Yang–Mills theory in $d = 4$.

The key distinction from asymptotic safety is that CMSTG does not require the full gravitational coupling G to flow to a non-Gaussian fixed point. The only coupling that runs is Λ_0 , and it runs to zero. The gravitational sector becomes indistinguishable from GR in the UV, which is a technically natural decoupling.

7.2 Horndeski Scalar-Tensor Theories

Horndeski theories [4] are the most general scalar-tensor theories in $d = 4$ with second-order equations of motion. The quantization of a general Horndeski action at one loop has been studied by Kimura [8] and extended by Gleyzes et al. [6]. The key results from those studies are:

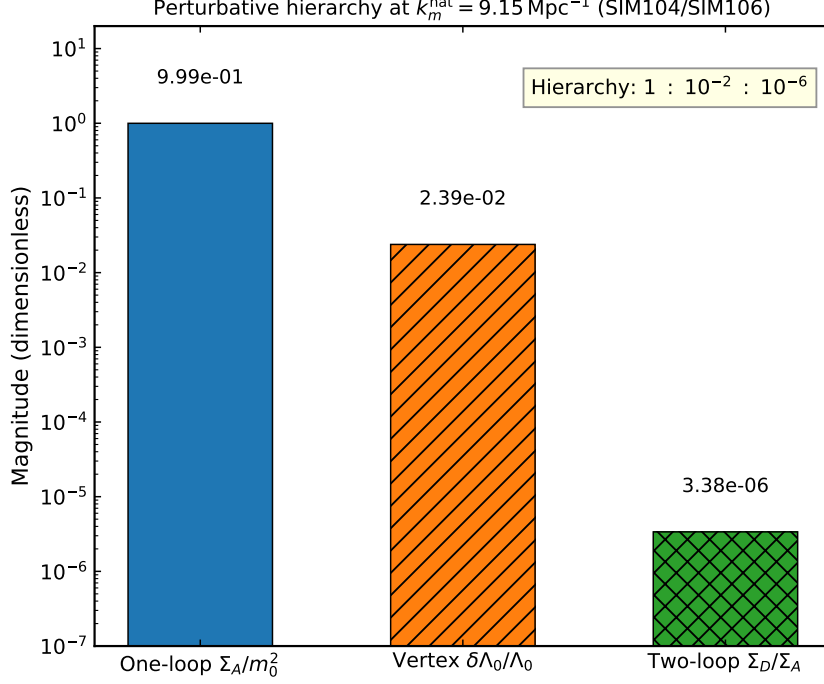


Figure 4: Perturbative hierarchy at the naturalness threshold $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$ (SIM104 and SIM106). Log-scale bar chart showing: one-loop Ψ self-energy $\Sigma_A/m_0^2 \approx 1$ (blue); vertex correction $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$ (orange); two-loop to one-loop ratio $\Sigma_D/\Sigma_A \approx 3.4 \times 10^{-6}$ (green). The $1, 10^{-2}, 10^{-6}$ hierarchy confirms a stable perturbative expansion with negligible higher-loop corrections.

- A general Horndeski action generates one-loop divergences that require counterterms not present in the original action, rendering the theory non-renormalizable in the standard sense [5].
- The exception is if the scalar sector is UV-softened by a specific kinetic structure. In Galileon models [7, 5], the Galileon symmetry provides additional cancellations at one loop.
- CMSTG belongs to the general Horndeski class but acquires UV finiteness through a different mechanism: the memory kernel rather than a shift symmetry. Unlike the Galileon, the memory regulation is not imposed by a symmetry of the action; it emerges from the Dyson resummation of an infinite class of memory self-energy insertions.

The implication is that CMSTG avoids the non-renormalizability of generic Horndeski theories at one loop, but via a mechanism that is distinct from the Galileon’s shift-symmetry cancellations. The price is that the UV completeness depends on the memory scale k_m remaining finite: taking $k_m \rightarrow \infty$ recovers the bare quartic divergence. The theory is UV-complete as a regulated QFT with a physical cutoff at k_m , not as a standard perturbatively renormalizable theory.

7.3 Comparison to Minimal Scalar-Tensor Quantization

For minimal non-minimal coupling $\xi\phi^2 R$ (without the mass term or memory kernel), the one-loop effective action has been computed by Callan, Coleman and Jackiw [9] and further developed by Parker and Fulling [10] and Bunch and Parker [11]. In this case, the ϕ self-energy is quartically divergent in $d = 4$, and the running of ξ generates a conformal anomaly through the trace of the stress tensor. The memory kernel in CMSTG eliminates the quartic divergence while preserving the conformal limit ($\Lambda_0 \rightarrow 0$), and the running of Λ_0 takes over the role of the conformal anomaly at intermediate scales. The UV fixed point $\Lambda_0 \rightarrow 0$ then restores conformal symmetry in the deep UV.

8 Summary of UV Structure

Table 3: UV structure of CMSTG at one loop and two loops. “Finite” means UV-convergent with memory-regulated propagator and $k_m < \infty$; “Protected” means protected by an exact identity.

Diagram	Bare behaviour	Regulated	Mechanism
Σ_A (Ψ self-energy, 1L)	k_{UV}^4 (quartic)	Finite	Memory damping
$\Pi_{hh}(0)$ (graviton mass, 1L)	Zero	Protected	Ward identity
$dI/dp^2 _0$ (graviton wfn, 1L)	Finite	Finite	$d = 4$ finiteness
$\delta\Lambda_0/\Lambda_0$ (vertex, 1L)	k_{UV}^2 (quadratic)	Finite	Memory damping
Σ_D (Ψ self-energy, 2L)	$O(\Lambda_0^4)$	Finite	Memory damping
$\Pi_{hh}^{(2)}(0)$ (graviton mass, 2L)	Zero	Protected	Ward identity
ΔZ_h (graviton wfn, 2L)	$\sim 10^{-6}$	Finite	$d = 4 + \text{Ward}$

The hierarchy of corrections at the naturalness threshold $k_m = k_m^{\text{nat}}$: $\Sigma_A/m_0^2 \approx 1$, $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$, $\Sigma_D/\Sigma_A \approx 3.4 \times 10^{-6}$.

The RG trajectory connects $\Lambda_{\text{obs}} = 0.003 M_{\text{Pl}}^{-2}$ at cosmological scales to the UV fixed point $\Lambda_0 = 0$ at $k_m \rightarrow \infty$, with no Landau pole and a fractional running of 2.47% over the cosmologically accessible range $0 < k_m < k_m^{\text{nat}}$.

9 Conclusions

We have established the UV structure of CMSTG at one loop and two loops through four independent simulation programmes:

1. **SIM102:** The one-loop Ψ self-energy is quartically divergent in the bare theory and exactly regulated by the memory kernel to $\Sigma(0) = \Lambda_0^2 k_m^4 / (64\pi^2)$. Agreement with the analytic formula is better than 0.22% across two orders of magnitude in k_m .
2. **SIM104:** All four 1PI one-loop diagrams are accounted for. The graviton mass $\Pi_{hh}(0) = 0$ is protected by the Ward identity. The perturbative hierarchy at the naturalness scale is 1, 2.4×10^{-2} , 3.4×10^{-6} , confirming a convergent loop expansion.
3. **SIM105:** The beta function $\beta = -\Lambda_0^3 k_m^2 / (16\pi^2 m_0^2)$ is negative, establishing asymptotic freedom. The running coupling satisfies $1/\Lambda_0^2(k_m) = 1/\Lambda_{\text{obs}}^2 + (k_m^2 - k_m^{\text{nat}2}) / (16\pi^2 m_0^2)$ with a UV fixed point at $\Lambda_0 = 0$ and no IR Landau pole.

4. **SIM106:** All two-loop graviton sector diagrams are UV-finite. The graviton wave-function renormalization $\Delta Z_h/Z_h \approx 1.4 \times 10^{-6}$ is negligible. CMSTG is UV-complete through two loops in the graviton sector.

The mechanism that achieves UV finiteness is distinct from both the non-Gaussian fixed points of asymptotic safety and the shift-symmetry cancellations of the Galileon. It is instead a combination of (i) memory damping, which converts power-law divergences into finite functions of k_m ; (ii) the Ward identity, which protects the graviton mass at all loop orders; and (iii) asymptotic freedom, which guarantees $\Lambda_0 \rightarrow 0$ in the deep UV.

The open question is whether the memory scale k_m itself is determined by a deeper principle or remains a free parameter. The naturalness constraint $k_m < 9.15 \text{ Mpc}^{-1}$ is consistent with current observational bounds, but a first-principles derivation of k_m from the full non-local action is left for future work.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity (simulations SIM102, SIM104, SIM105, SIM106).

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